

Fundamental concept of boolean algebra

Only two mathematical operations

Addition

↑
OR
↑

Conjunction (logical addition)

The '+' symbol
is read as
x or y

$$\begin{aligned} x + y \\ 0 + 0 &= 0 \\ 0 + 1 &= 1 \\ 1 + 0 &= 1 \\ 1 + 1 &= 1 \end{aligned}$$

Multiplication (Repeated addition)

↑
AND
↑

Disjunction (Logical multiplication)

$x \cdot y$ The dot symbol
is read as x and y

$$\begin{aligned} x \cdot y \\ 0 \cdot 0 &= 0 \\ 0 \cdot 1 &= 0 \\ 1 \cdot 0 &= 0 \\ 1 \cdot 1 &= 1 \end{aligned}$$

1 and 0,
high and low,
true and false,
on and off.

George Boole, has written a book "An Investigation of the laws of thought". This book is based on logic and Probabilities.

Later on Claude E. Shannon at Massachusetts Institute of Technology wrote a thesis. Today Boolean Algebra is the back bone of Computer Circuit analysis.

Three basic laws of Boolean Algebra are same as normal algebra.

1. Commutative law:- $A + B = B + A$, $A \cdot B = B \cdot A$

2. Associative law: $(A + B) + C = A + (B + C)$, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

3. Distributive law: i) $A(B + C) = AB + AC$
ii) $A + B \cdot C = (A + B)(A + C)$

x } = variables
 y } = stores decimal
numbers in
normal algebra.
But in boolean
algebra it will
store only
binary values

Logic Gates: A logic gate is defined as an electronic circuit

store only binary values

Logic Gates: A logic gate is defined as an electronic circuit with two or more input signals and have one output signal and it depicts the logic clearly.

There are 3 types of logic Gates :

1. Basic Gates

→ OR gate
→ AND gate
→ NOT gate

2. Universal Gates

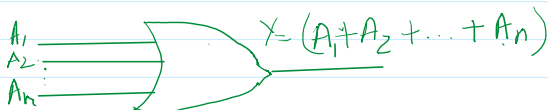
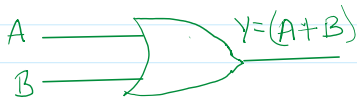
→ NOR gate
→ NAND gate

3. Exclusive Gates

→ XOR gate
→ XNOR gate

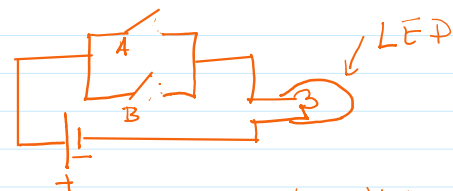
BASIC GATES

1. OR Gate :- An OR Gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on; if any of the input signal is high or 1 or true or on. This is called (logic-1) gate.



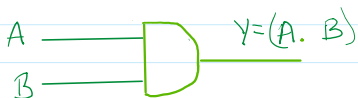
	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$2^n = 2^2 = 4$
combinations
(0, 1, 2, 3)



Parallel switch connection

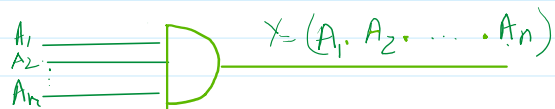
ii) AND Gate :- An AND Gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on; if all of the input signal is high or 1 or true or on. The logic is represented by dot '.' symbol.



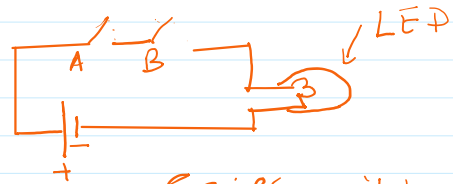
	A	B	Y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$2^n = 2^2 = 4$
combinations
(0, 1, 2, 3)

c

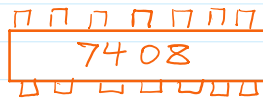


3 1 1 1

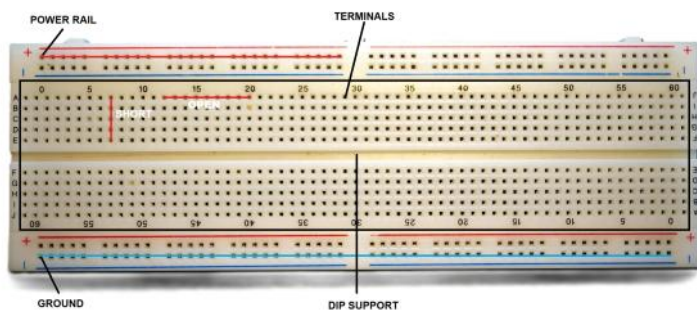
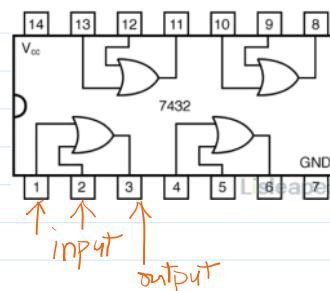
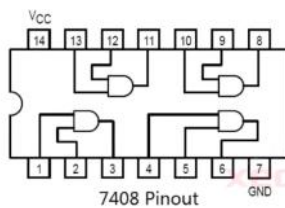
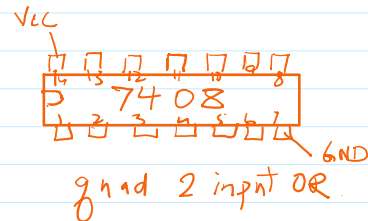


Series switch connection

Integrated ckt. (IC)



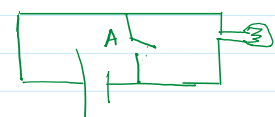
Quad 2 input AND



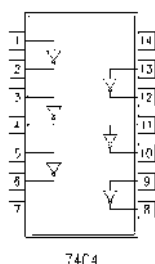
NOT Gate $A \rightarrow \overline{A}$

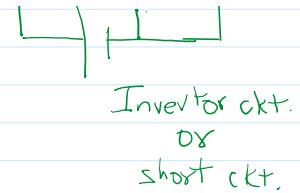
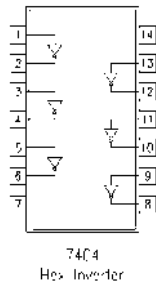
An inverter or NOT gate is a ckt. that takes one input signal and output signal is always opposite or complement of the input signal.

A	$\overline{A}$
0	1
1	0



Inverter ckt.
OR
short ckt.





NOT GATE IC

BASIC DUALITY IN BOOLEAN ALGEBRA

We state the duality theorem in the following:

1. Changing each OR (+) sign to an AND (.) sign.
2. Changing each AND (.) sign to an OR (+) sign.
3. Complement each '0' to '1' or '1' to '0'.

$$A + 0 = A \cdot 1$$

$$A + B \text{ dual is } A \cdot B$$

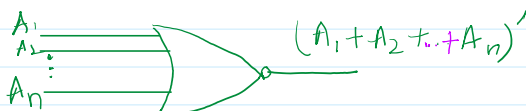
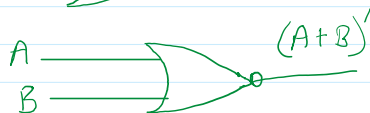
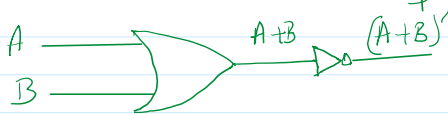
$$\text{dual} \rightarrow \left\{ \begin{array}{l} A \cdot (B + C) = AB + AC \\ (A + B) \cdot C = (A + B)(A + C) \end{array} \right\} \text{ distributive law relationship}$$

$$1 + A \text{ dual is } 0 \cdot A$$

$$1 + \bar{A} \text{ dual is } 0 \cdot \bar{A}$$

UNIVERSAL Gates (De-Morgan's law)

1. NOR gate: This gate takes two or more input and output is inverter of OR gate.



A	B	(A+B)'
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table of NOR gate

De-Morgan's law

$$(A+B)' = \bar{A} \cdot \bar{B}$$

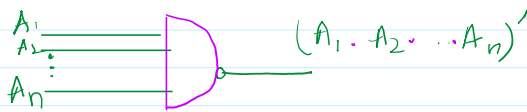
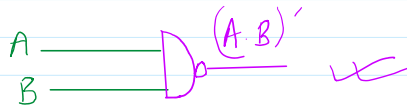
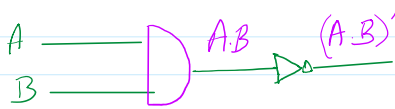
1. dual. of this expression

De-Morgan's law $\rightarrow (A+B)' = \bar{A} \cdot \bar{B}$

$\downarrow$ dual of this expression

$(A \cdot B)' = \bar{A} + \bar{B}$

1. **NAND** Gate: This gate takes two or more input and output is inverter of AND gate.



A	B	$(A \cdot B)'$
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of NOR gate

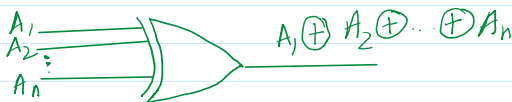
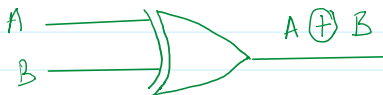
$$(A \cdot B)' = \bar{A} + \bar{B}$$

EXCLUSIVE GATES

1. **XOR** Gate: It takes 2 or more input signals and output one signal.

The output signal is high or/ or true, iff we had odd number of high or/ or true in a row of input signals, otherwise output is low or 0 or false.

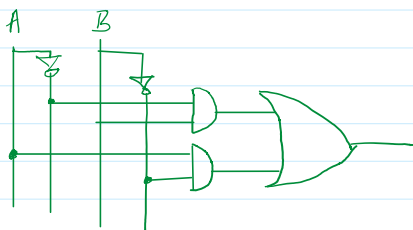
It is represented by $\oplus$ plus in a circle symbol.



A	B	$A \oplus B$	
0	0	0	$\leftarrow A + B$ (Product of sum)
0	1	1	$\leftarrow \bar{A} B$ (SOP)
1	0	1	$\leftarrow A \bar{B}$ (SOP)
1	1	0	$\leftarrow \bar{A} + \bar{B}$ (Product of sum)

POS $\rightarrow A \oplus B = (A+B) \cdot (\bar{A} + \bar{B})$

SOP $\rightarrow A \oplus B = \bar{A} B + A \bar{B}$



XOR Gate using basic gates.

HW: Draw the logic gate ckt. for XOR gate using the expression POS.

XOR gate using basic gates.

SOP Expression.

	A	B	C	$A \oplus B \oplus C$
0	0	0	0	0 ✓
1	0	0	1	1 ✓
2	0	1	0	1 ✓
3	0	1	1	0 ✓
4	1	0	0	1 ✓
5	1	0	1	0 ✓
6	1	1	0	0
7	1	1	1	1 ✓

1 is selected as output
0 is barred/complemented
SOP

$$A \oplus B \oplus C = \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC$$

POS ← 1 is barred/complement
0 is selected as output

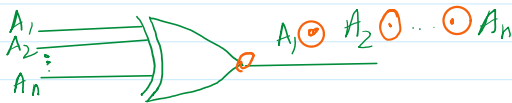
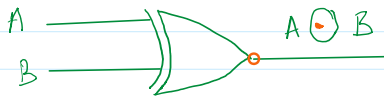
$$A \oplus B \oplus C = (A+B+C)(A+\bar{B}+\bar{C})(\bar{A}+B+\bar{C})(\bar{A}+\bar{B}+C)$$

HW. Draw the logic ckt. using basic gates of SOP & POS.

1. XNOR gate: It takes 2 or more input signals and output one signal.

The output signal is high or 1 or true, iff we had **even** number of high or 1 or true in a row of input signals, otherwise output is low or 0 or false.

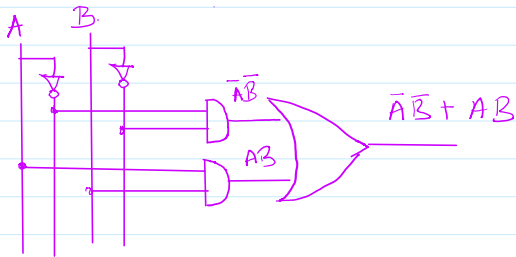
It is represented by '⊙' dot in a circle symbol.



A	B	$A \odot B$	
0	0	1	$\bar{A} \cdot \bar{B}$ (Sum of Product) SOP
0	1	0	$A + \bar{B}$ POS
1	0	0	$\bar{A} + B$ (Product of sum)
1	1	1	AB

SOP → $A \odot B = \bar{A}\bar{B} + AB$

POS → $A \odot B = (A+\bar{B})(\bar{A}+B)$



HW: Draw the logic gate ckt. for XNOR gate using the expression POS.

XNOR gate using basic gates.

XNOR gate using basic gates.

SOP Expression.

	A	B	C	A ⊕ B ⊕ C
0	0	0	0	1
1	0	0	1	0
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

1 is selected as output
0 is barred/complemented
SOP

$$A \oplus B \oplus C = \bar{A}\bar{B}\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

PDS ← 1 is barred / complement
0 is selected as output

$$A \oplus B \oplus C = (A+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(\bar{A}+\bar{B}+\bar{C})$$

HW. Draw the logicalckt. using basic gates of SOP & PDS.

Boolean Algebraic Laws

1. Logical Addition(OR)

- i. $0 + 0 = 0$
- ii. $0 + 1 = 1$
- iii. $1 + 0 = 1$
- iv. $1 + 1 = 1$

2. Logical Multiplication

- i. $0.0 = 0$
- ii. $0.1 = 0$
- iii. $1.0 = 0$
- iv. $1.1 = 1$

3. Complement of bits

- i. $0' = 1$
- ii. $1' = 0$

4. Laws of boolean algebra

Law	Properties
$0 + p = p$	0
$0.p = p$	
$1 + p = 1$	1
$1.p = p$	
$p + p = p$ $p.p = p$	Idempotence law
$p + p.q = p$ $p + (p + q) = p + q$	Absorption law
$(p')' = p$	Involution law
$p + p' = 1$ $p.p' = 0$	Complementarity law
$(p + q) + r = p + (q + r)$ $(p.q).r = p.(q.r)$	Associative law

$p(q + r) = p.q + p.r$ $p + q.r = (p + q)(p + r)$ $p + p'.q = p + q$	Distributive law
$(p + q)' = p'.q'$ $(p.q)' = p' + q'$	Demorgan's law
$p \Rightarrow q = p' + q$	Implication or Conditional Elimination or if...then
$(p \Leftrightarrow q) = (p \Rightarrow q).(q \Rightarrow p) = p'.q' + pq$	Bi-conditional elimination or equivalence or if and only if (iff)

p	q	p'	q'	(p+q)'	p'.q'	(p.q)'	p' + q'
0	0	1	1	1	1	1	1
0	1	1	0	0	0	1	1
1	0	0	1	0	0	1	1
1	1	0	0	0	0	0	0

$\uparrow \quad \uparrow$
 $(p+q)' = p'.q'$ $\uparrow \quad \uparrow$
 $(p.q)' = p' + q'$

What is truth table?
 A truth table is a systematic way to represent all possible combination of inputs and their corresponding outputs for a logical expression or circuit. It is widely used in Boolean Algebra, digital logic and design, and computer science to analyze and validate logical operations.

{ NAND to NAND
 or
 NOR to NOR logic.